KONAMARTHI NITISH VEERA VENKATESH

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Summary

Computer Science & Engineering graduate and AWS Certified AI Practitioner with hands-on experience building LLM-powered applications in Python, integrating REST APIs, and applying Generative AI, RAG, embeddings, vector databases, and prompt engineering concepts. Experienced in deploying AI solutions on cloud platforms (AWS, OCI) with Git-based workflows; currently learning LangChain.

Education

Anil Neerukonda Institute of Technology and Sciences Sep 2022 – Jun 2026
B.Tech in Computer Science & Engineering *Visakhapatnam, India*

CGPA: 8.92/10.0 | **Coursework:** Data Structures & Algorithms, OOP, DBMS, Software Engineering, Cloud Computing

Certifications

- **AWS Certified AI Practitioner (AIF-C01)** – Generative AI, RAG, embeddings, vector databases [Badge]
- **OCI 2025 Certified AI Foundations Associate** [Badge]
- **AWS Academy Graduate – Cloud Foundations** [Badge]
- **OCI 2025 Certified Foundations Associate** [Badge]
- **Postman API Fundamentals Student Expert** [Badge]

Technical Skills

Programming & Data: Python, SQL (MySQL, PostgreSQL), JSON, Java, JavaScript, TypeScript
Generative AI: LLMs, Prompt Engineering, RAG, Embeddings, Vector Databases, LangChain (learning), Google Gemini API
Machine Learning: Deep Learning, Image Classification, DenseNet-121, Model Evaluation
APIs & Backend: REST APIs, FastAPI, Node.js, Fastify, Express.js, Postman
Cloud: AWS (Bedrock, EC2, S3, Lambda, VPC), Oracle Cloud Infrastructure (OCI)
Tools: Git, GitHub, Linux, VS Code

Projects

Chest X-Ray Diagnostic Assistant *Python, FastAPI, DenseNet-121, Gemini API, Geoapify*

- Developed a DenseNet-121 deep learning model to detect and classify 14 chest X-ray pathologies, achieving 87.8% accuracy.
- Built a Python FastAPI REST inference backend and integrated the Google Gemini LLM API with prompt engineering to convert model predictions into structured, explainable reports.
- Implemented image validation and preprocessing ahead of inference to reject invalid inputs and improve output reliability.
- Deployed on Vercel and Hugging Face Spaces; integrated the Geoapify API to recommend nearby healthcare providers.

TokenSentry – AI Governance & Model Routing Layer for Coding Agents Expected Completion: Q4 2026
Node.js, Fastify, TypeScript, LLM APIs *In Progress*

- Building a plugin-based proxy layer between coding agents and upstream LLM APIs that intercepts requests to enforce usage policies and token budgets without changing the agent's workflow.
- Designing a cost-aware model routing engine that selects an LLM per request using provider, capability, and pricing metadata.
- Architecting a provider-agnostic design that separates governance logic from provider connectivity and API-key handling.

Achievements

- **Smart India Hackathon 2024:** Selected at national level for problem SIH1637 (Agriculture & Rural Development); collaborated with a cross-functional team to build an agri-tech prototype.
- **ANITS x PivotalSoft 24-Hour Hackathon 2024:** Built a full-stack prototype within 24 hours, demonstrating rapid prototyping and problem-solving.